

Isha Ubale

858-257-8375 | iubale@ucsd.edu | [linkedin.com/in/isha-ubale](https://www.linkedin.com/in/isha-ubale)

EDUCATION

University of California San Diego

B.S. Bioengineering: Biotechnology

Sep 2022 – Jun 2026

San Diego, CA

EXPERIENCES

Bowman Neurobiology Imaging Lab, Salk Institute

Lab Technician

Sep 2025 – Present

San Diego, CA

- Supported EO-FLIM experiments from sample prep to acquisition; assisted imaging QC to maintain data quality.
- Performed Drosophila husbandry and genetic crosses to generate experimental lines for imaging while maintaining laboratory stock record.
- Prototyped and iterated Fusion 360 CAD surgical mounts, improving alignment precision and reducing microsurgery prep time by 20%, increasing experimental throughput.

Cartilage Tissue Engineering Lab, UCSD

Undergraduate Researcher

Jun 2024 – Apr 2025

San Diego, CA

- Helped develop and validate a quantitative Ca^{2+} spectrophotometric assay to enable reliable bone mineral measurements, improving accuracy and reproducibility across samples.
- Optimized oCPC-reagent dilution ranges and built calibration curves to expand the assay's usable range and improve measurement consistency.
- Prepared bone samples for testing and executed data collection with organized documentation to support reproducible analysis.

PROJECTS

Flow-Based Organoid Sorting Device — Capstone Senior Design

Aug 2025 – Present

- Converted clinical and research needs into measurable device specifications to guide microfluidic design decisions and verification planning.
- Established morphology-based pass/fail criteria for sorting and outlined the image features needed to support classification.

Wearable Air Quality Monitor — Engineering World Health

Oct 2024 – May 2025

- Contributed to an Arduino-based wearable that alerts users via vibration/LED when pollutant thresholds are exceeded in local air.
- Supported sensor integration and threshold logic; participated in iterative prototyping to improve reliability and user-facing feedback.

LEADERSHIP

Project Lead — Engineering World Health

Oct 2025 – Present

- Led early-stage project requirements definition and translated user needs into measurable prototype performance goals.
- Guided and documented materials research and selection and outlined validation plans.
- Coordinated team execution through task assignment and milestone tracking to maintain momentum.

Instructional Assistant — Introduction to Bioengineering, UCSD

Jan 2024 – Mar 2024

- Supervised 2 lab teams of 5–8 undergraduate students during weekly sessions, ensuring proper lab execution and troubleshooting in real time.
- Prepared instructional materials including protocols, quizzes, and lab station setup to improve clarity and learning outcomes.
- Demonstrated use of lab equipment (micropipettes, breadboards, calipers) and reinforced proper use.

CORE SKILLS

- **Engineering:** Autodesk Fusion 360 (CAD), 3D printing, Arduino.
- **Computational:** MATLAB, Python, ImageJ, Figma.
- **Communication:** Technical writing, research presentations, interdisciplinary teamwork.